MES COLLEGE OF ENGINEERING-KUTTIPPURAM
DEPARTMENT OF COMPUTER APPLICATIONS
20MCA246 – MAIN PROJECT

PRO FORMA FOR THE APPROVAL OF THE FINAL SEMESTER PROJECT

(Note: All entries of the pro forma of approval should be filled up with appropriate and complete information. Incomplete Pro forma of approval in any respect will be rejected.)

Project Proposal Number : 1
(Filled by the Department)

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Academic Year : 2024-2026
Year of Admission : 2024
Admission Number : 18931
Roll Number : 45
Register Number : MES24MCA-2045

1. Name of the Student (in BLOCK LETTERS) : RIZVANA K A
2. Name of the Organization : ____
3. Address of the Organization : ____
   Telephone No. : ____ Company E-Mail : ____
4. Name of the External Guide : ____
   Mobile No. : ____ E-Mail : ____
5. Title of the Project : AI BASED FLAT SECURITY
6. Name of the Guide (Internal-Departmet) : BALACHANDRAN K P

Date : 5-1-2026 Signature of the Student:

Comments of The Project Guide

Initial Submission :

Approval Status : Approved / Not Approved Dated Signature of Guide HOD

First Review :

Second Review :

Third Review :

Comments of The Project Coordinator

Initial Submission:

First Review : Second Review: Third Review:

Dated Signature of Project Coordinator:

AI Based Flat Security

Abstract

The AI-Based Flat Security System is developed to improve safety in residential apartments by using Artificial Intelligence technology. With the increase in apartment living, security issues such as unauthorized entry and delayed emergency response have become common. Traditional security systems mainly depend on security guards and CCTV cameras, which require constant monitoring and are not always effective.

This system uses camera-based surveillance along with AI face recognition to identify people entering the apartment. The captured face images are compared with stored data of authorized residents. Based on this comparison, the system identifies whether the person is a known resident or an unknown visitor.

When an unknown person is detected, the system immediately sends alerts to the flat owner and security staff. This allows quick action to be taken and helps prevent possible security threats. The system also stores visitor entry and exit details for future reference.

In addition, the system provides a QR-code-based visitor management feature. Residents can generate time-limited QR codes for their guests, making the entry process secure and easy to track. Other features such as emergency alerts, complaint registration, and notifications are managed through a single digital platform.

Overall, the AI-Based Flat Security System offers a simple, smart, and efficient solution for apartment security. It reduces human effort, improves response time, and creates a safer and more organized living environment by using modern AI technology.